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ORIGINAL ARTICLE

Dynamic AI-Embedded Super App: A Design-Based Process Innovation for Customer Engagement and Value Creation

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ABSTRACT

Academic: In an era where mobile applications (apps) struggle for longevity amidst fluctuating download and deletion rates, the emergence of super apps marks a pivotal transformation. A super app is a mobile app-based service aggregator platform that promises extensive value for customers and businesses. While early efforts like Apple's MobileMe and Google Buzz yielded mixed results, generative AI technology now positions super apps as a viable, all-encompassing solution. This study explores the rise of the super app phenomenon and its potential to revolutionize mobile engagement and usage. Underpinned by the human-centered design-thinking approach, comprising desirability, feasibility, and viability elements, this study defines the super app as a multi-dimensional phenomenon and further identifies the key elements needed to develop a successful super app. The study proposes critical factors and a process framework for the successful development and adoption of super apps. The findings reveal that users' market context (emerging vs. developed) and network co-dependence act as important moderating factors influencing the process of super app adoption and customer engagement. The study advances the domain of innovation management and proposes a strategic matrix for managers to plan and align stakeholder engagement activities effectively. The study opens avenues for future research in engagement, value creation, and digital innovation.

Managerial: In today's crowded app market, most apps have short lives and uneven customer engagement. Super apps mark a shift by combining services such as payments, e-commerce, and communication in a single platform. They use adaptive interfaces and personalized features to keep users active. With generative AI improving recommendations and support, super apps drive stronger engagement and deliver value to customers and businesses. This study conceptualizes super app as an AI-embedded, multi-dimensional phenomenon grounded in human-centered design thinking, emphasizing the critical role of elements of desirability, feasibility, and viability. It identifies various key customer-level, firm-level, and context-level elements for building successful super apps. Findings emphasize that the adoption of super apps and customer engagement vary across markets (emerging vs. developed economies) and are based on the degree of network co-dependence. To support implementation, the study introduces a 2×3 strategic matrix that links market penetration levels (high, moderate, low) with stakeholder engagement (customers and participating firms). Suggested strategies include sustenance (e.g., AI-powered loyalty programs), growth (flexible revenue-sharing models), and acquisition (service bundling and innovation). Finally, while metadata collection enables sharper insights, managers must ensure strong data privacy, encryption, and monitoring to protect trust for sustaining long-term customer and ecosystem relationships.

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1 | Introduction

A Gartner report predicts that more than half of the global population will become daily users of super apps on their mobiles by 2027, transforming both mobile and desktop experiences through generative artificial intelligence (AI) (Gartner 2022). Coined by Mike Lazaridis, the founder of Blackberry in 2010, the term 'super app' refers to a unified digital ecosystem that provides access to multiple services with a single, seamless, and integrated interface (Robb 2010).

A super app[1] is a comprehensive mobile platform that consolidates a range of services by integrating offerings from various participating firms into one cohesive system. Participating firms, comprising service providers and mini apps, leverage the super app ecosystem to efficiently reach users and offer services. Some globally recognized super apps enable users to access services from multiple providers through their apps. For example, WeChat provides a seamless interface that integrates utilitarian services (e.g., mobile payments, e-commerce, transportation) with hedonic ones (e.g., messaging, social networking, and entertainment).

The digital sphere has increasingly become an extension of global consumers' lives. This trend is particularly pronounced in emerging markets[2] where rapid mobile phone adoption has fueled the proliferation of mobile applications. However, mobile phone users in these regions often face challenges managing multiple applications for different services. Similarly, app firms struggle with inconsistent user engagement, low retention rates, and high churn levels. Research suggests that 80%–90% of downloaded apps are uninstalled after a single use, with the average uninstallation time being just 6 days (Varshney 2020). Such high churn rates threaten the long-term viability of mobile app firms as they struggle to maintain user engagement and platform stickiness. As a result, the need for a more streamlined and efficient approach to digital service consumption has never been more critical, an issue that super apps aim to address.

A super app is a mobile platform that integrates services from multiple providers into one seamless ecosystem, enabling services from payments to messaging (Flora et al. 2014). Unlike traditional standalone apps or multi-purpose app suites, super apps operate as open ecosystems, reducing app clutter and enhancing user experience. These differences extend to key aspects such as user interface, scalability, and value creation (see Web Appendix—A1).

Though marketplace evidence highlights successful super apps (e.g., WeChat) and unsuccessful attempts (e.g., LINE, Uber), critical inquiries arise: What determines a super app's success in adoption and user engagement? What role does consumer demand play in driving this success? What essential design and investment factors must firms prioritize to ensure success? What role does generative AI play in advancing super apps' design, operational efficiency, and functionality? How may the contextual or behavioral factors may moderate the way users engage with super apps and how firms derive value from them? Existing literature remains silent in exploring the factors underlying their success or failure. Addressing these inquiries can unlock the value potential super apps can generate for key stakeholders (customers, super app firms, and participant firms).

Hence, the research objectives of this study are as follows: (1) To identify the pivotal market dynamics and forces compelling the strong desire for super apps among firms and consumers; (2) to explore the critical elements firms should consider while designing and investing in developing a successful super app; (3) to identify the key characteristics that define a super app; (4) to investigate the role generative AI might play in designing and operationalizing a super app; and (5) considering the potential moderating conditions, to find out how app firms can expedite user engagement and derive value for the stakeholders. While investigating these research objectives, this study conceptualizes super apps as a human-centric design innovation that amplifies smartphone utility, employing the design thinking approach (Chasanidou et al. 2015; Seidel and Fixson 2013). Additionally, the study draws on customer engagement theory to frame how users interact with super apps and to uncover mechanisms through which engagement can be deepened to drive long-term value (Kumar and Pansari 2016; Venkatesan 2017).

This research significantly enhances the understanding and development of super apps within the dynamic digital ecosystem in three ways. First, it offers a conceptualization of the super app as a multi-dimensional phenomenon, distinguishing it from conventional mobile apps by highlighting its capacity to deliver a unified, seamless experience across diverse services (Mekuria and Mfupe 2017; Shankar and Narang 2020; Watson and Sauter 2011). Second, it identifies the key factors contributing to super app success and proposes a structured process framework rooted in design thinking to enhance customer engagement and multi-stakeholder value creation (Kumar and Pansari 2016; Pansari and Kumar 2017; Venkatesan 2017). By placing user needs at the center and optimizing engagement mechanisms, the study demonstrates how super apps can deliver increased value for consumers and businesses (Meinel et al. 2020). Finally, it outlines practical implications and strategic insights for managers, while opening avenues for future research on how super apps can be leveraged to build and sustain competitive advantage as an innovation.

2 | Triangulation Approach

This study employs a triangulation approach, a well-established method in qualitative research that ensures construct validity by integrating data from multiple sources (Holt and Thorpe 2007; Patton 1999). This approach strengthens the robustness of findings by cross-verifying data, thereby reducing bias and enhancing the credibility of the results.

The triangulated insights are generated from the literature review, marketplace evidence, and interviews to examine the adoption and design of AI-embedded super apps. First, we review the academic literature on digital platforms, innovation, and AI adoption, noting the lack of conceptual clarity and limited studies on super apps. Second, we analyze successful and

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failed attempts in the market (e.g., WeChat, LINE, Uber) to understand platform strategies, integration models, and user uptake patterns. Finally, we draw on interviews with customers and industry stakeholders (super apps and participating apps) to uncover factors essential for understanding the designing, adoption, and scaling of super apps in digitally converging ecosystems.

2.1 | Literature Review

Super apps represent a major shift in the mobile app ecosystem and have attracted growing interest across both industry and academia. Existing literature reflects the complexity of super app development and adoption, with fragmented studies scattered across multiple disciplines such as information systems, digital strategy, and finance (Kavitha et al. 2023). However, despite this growing body of work, there remains a lack of integrated understanding of the factors that drive super app success. To address this gap, Table 1 presents a curated selection of studies: the first part focuses on key market contexts and development drivers, while the second outlines defining characteristics of super apps. A more detailed review can be found in Web Appendix—A2.

One of the main drivers of super app adoption is the limitation of traditional mobile apps in addressing diverse user needs, especially in the context of limited smartphone storage (Cheng et al. 2020). Super apps offer an integrated platform with mini-app ecosystems, seamless user interfaces, and enhanced usability (Nayebi and Ruhe 2017; Roa et al. 2021). Other important design considerations include privacy, discoverability, scalability, and security (Cheng et al. 2020; Ota et al. 2020). Despite these common traits, the literature lacks a universally accepted definition of a super app (Zhu et al. 2023). This conceptual ambiguity has made it difficult to identify clear criteria for distinguishing super apps from conventional mobile applications (Chen et al. 2018).

While most research focuses on functionality and market contexts, there is growing interest in how super apps shape consumers' behavior and wider social outcomes. For instance, platforms like Alipay and WeChat have transformed fintech and payment systems by promoting financial inclusion and frictionless transactions (Fasnacht 2021; Kavitha et al. 2023; Roa et al. 2021). Others, such as LINE and Grab, are seen to have a broader social and political impact, embedding themselves in daily routines and public discourse (Huang and Miao 2021; Steinberg 2020).

A key concept emerging in this literature is network co-dependence, the idea that social connections among users reinforce app adoption and engagement (Markus 1987; McPherson et al. 2001; Salehi et al. 2023). Peer influence, particularly in emerging markets, makes these apps central to everyday life (Kumar et al. 2019). In such contexts, super apps often compensate for weak infrastructure, helping users bypass fragmented service delivery.

Adoption patterns differ significantly between emerging and developed markets. In emerging economies like India, China, and Indonesia, affordable smartphones and low-cost data plans have encouraged mobile-first usage. Here, super apps serve as essential gateways to digital services (Mekuria and Mfupe 2017). In contrast, users in developed markets typically rely on standalone, specialized apps. Stronger regulatory environments and existing infrastructure reduce the demand for bundled platforms (Shankar and Narang 2020).

One area that remains underexplored is customer engagement. This theory defines engagement as a cognitive, emotional, and behavioral investment in user-brand interaction (Dinner et al. 2015; Kumar and Pansari 2016). While widely applied to digital platforms, it has rarely been used to study super apps. This is surprising, given that super apps aim to foster habitual and emotionally resonant user experiences by integrating daily services (Venkatesan 2017). Our study addresses this gap by applying customer engagement theory to understand how super apps build deeper relationships and long-term value for the stakeholders.

Some studies also draw attention to potential risks. These include data privacy concerns, monopolistic behavior, and the concentration of user data within a single platform (Fasnacht 2021; Steinberg 2020; Zhu et al. 2023). On the other hand, researchers have also noted super apps' benefits to small retailers and local economies, primarily through expanded market access and digital storefronts (Roa et al. 2021). These contrasting perspectives highlight the need for more balanced research into the trade-offs super apps present for different stakeholders.

In summary, the literature reveals the promise and challenges of super apps. However, more profound conceptual clarity and stronger theoretical grounding, particularly in customer engagement and innovation, are still needed. Next, we explore marketplace practices and industry evidence to gain further relevant insights in this discussion.

2.2 | Marketplace Evidence

The evolving marketplace provides a rich set of real-world examples illustrating the mixed outcomes of super app development and adoption across different regions. By examining both successful and failed super app initiatives, we can better understand the conditions under which these platforms thrive or falter. This section draws on marketplace evidence to analyze the strategic, technological, and behavioral factors that have shaped the performance of prominent super apps across domestic and international markets.

Super apps like Apple MobileMe, Google Buzz, and Tata Neu failed to gain market traction (Business Insider 2010; CNET 2012; Inc42 2023). Their struggles can be attributed to multiple, recurring challenges. First, these apps were launched ahead of user readiness, offering multi-functional interfaces that overwhelmed users or demanded excessive personal information. Second, ambiguous positioning and unclear feature sets left users uncertain about the app's core value. Third, issues such as technical glitches, limited daily-use features, poor user experience, and ineffective marketing strategies hampered adoption. Finally, regulatory hurdles and privacy concerns eroded user trust, especially when apps required sharing sensitive data across partner services. These cases highlight

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TABLE 1 | Select literature review on super app and proposed super app characteristics.

Section 1: Literature review findings						Section 2: Super app characteristics					
Authors	**Context of study**	**Method**	**Market context**	**Factors for super app development**	**Findings**	**Integrated digital payment mechanism**	**Mediation platform**	**Inclusivity**	**Multi-purpose single platform**	**Individual app environment**	**Optimized functionality**
Nayebi and Ruhe 2017	Optimized super app functions	Exploratory study	Global	– Cohesiveness – Feature extraction – Effort estimation – User feedback	– An app is considered valuable when features are cohesive-co-occurrence of existing apps. – Cohesiveness ensures semantic connections across features are maintained and optimized.			✓	✓	✓	✓
Cheng et al. 2020	Super app innovation and mobile platform ecosystem	Exploratory study	Emerging market (China)	– Customer access – Distribution – Portability – Hardware/ software – Internet access	– The native apps do not meet the specific needs of users; can be supplemented by mini-programs. – Super app fills gaps in traditional apps by reducing fragmentation and app development cost	✓	✓		✓	✓	✓
Huang and Miao 2021	Micro-behavioral process of super app	Descriptive study	Emerging market (China)	– Re-domestication of technology – User privacy – Culture	– Customers suspend the use of the super app periodically, lacking engagement—Super app experience gets affected by prevailing social norms.		✓	✓	✓	✓	✓

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TABLE 1 | (Continued)

Section 1: Literature review findings						Section 2: Super app characteristics					
Authors	**Context of study**	**Method**	**Market context**	**Factors for super app development**	**Findings**	**Integrated digital payment mechanism**	**Mediation platform**	**Inclusivity**	**Multi-purpose single platform**	**Individual app environment**	**Optimized functionality**
Roa et al. 2021	Impact of alternative data from super app	Quantitative study	Developed market (Latin America)	– Customer demographics – Financial services – Type of data	Super-app diversity is attributable to its mini-program function—super app can be quite helpful for financial institutions	✓	✓		✓	✓	✓
Steinberg 2020	Cultural platformization of super app	Descriptive study	Emerging market (East Asia)	– Platformization – Geo-culture specific – Revenue model – Mediation	– Super apps encourage new producers to reach new markets. – The value of a culturally customized super app is a mutual dialogue with customers.	✓	✓	✓	✓		
Present Study	Design thinking AI- super app customer engagement	Triangulation method with thematic analysis	Emerging markets	– Viability – Desirability – Feasibility – Moderators – Value Creation	– Conceptualizing super app – Understanding the driving forces and factors for designing a super app, moderating factors influencing customer engagement, and deriving value for the stakeholders.	✓	✓	✓	✓	✓	✓

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the importance of clear value propositions, intuitive design, strong privacy safeguards, and consistent user trust-building.

In contrast, the success of super apps like WeChat in China and Gojek in Indonesia demonstrates their potential to dominate the mobile app space, particularly in their home markets. Marketplace evidence yields six key success factors: First, leading super apps leverage generative AI to deliver personalized experiences, secure payments, and optimized multi-purpose mini-apps. For example, WeChat uses generative AI for customized notifications and data security enhancements. Second, strategic partnerships with tech-driven firms enhance app efficiency and marketing. For instance, Gojek partnered with TenMax.io, a digital advertising company in Taiwan, to deploy generative AI-powered advertising and optimize digital reach (Marketing-interactive 2021). Third, a firm's financial strength and technological resources are crucial to super app success (Roa et al. 2021). Super apps such as WeChat, Gojek, Alipay, and KakaoTalk have succeeded partly due to strong funding, advanced tech infrastructure, and brand credibility. These attributes helped build user trust in privacy and data security, facilitating deeper engagement.

Fourth, super apps tend to thrive in emerging markets, where users often bypass desktops and adopt mobile-first behaviors, a phenomenon known as mobile leapfrogging (Shankar and Narang 2020; Watson and Sauter 2011). Affordable, low-storage smartphones and expanding internet access have enabled rapid digital inclusion. The Asia-Pacific region, led by WeChat in China, has seen strong growth fueled by AI infrastructure and tech talent. However, even dominant apps face globalization challenges. Despite its domestic success, WeChat struggled internationally due to its inability to adapt to the cultural preferences and market expectations (Fasnacht 2021).

Fifth, market localization is a major driver of adoption. Native super apps, such as Go-Viet (Vietnam), Kakao (South Korea), Go-Jek (Indonesia), Paytm, and Jio (India), benefit from an in-depth understanding of local needs and behaviors. Their country-specific focus allows them to tailor features and services more effectively than their global counterparts.

Sixth, aligning services with local lifestyles and cultural contexts boosts customer engagement. For instance, Careem, a Dubai-based super app, personalizes the ride-hailing experience by adjusting arrival times based on weather and religious prayer schedules (McKinsey 2022).

Together, these insights provide a multi-dimensional understanding of the factors that shape super app success and value. Building on these findings, we interviewed customers, super app managers, and partner firm executives to capture firsthand perspectives (Table 2).

2.3 | Stakeholders Interviews

We conducted semi-structured interviews with diverse stakeholder groups to understand user needs, adoption factors, and app interaction behaviors from a grounded perspective. Participants were recruited through purposive and snowball sampling, starting with known customers of digital payment and mobile platforms, managers, and executive heads working in super app platforms or mini app firms, followed by referrals across age groups and user profiles.

The following criteria were considered when choosing the stakeholder respondents: *Customers* were chosen based on (1) their daily usage of various mobile apps, (2) the amount of usage of mobile wallets and UPIs for payment and transactions, (3) excitement or inertia towards the discussion of mobile apps, and (4) the value and relevance they see in mobile apps usage over the traditional way of transacting. Respondents are from various age groups: 18–30, 30–45, and 45 and above. Interviews were conducted in a hybrid format, that is, in-person, phone, and online conversations, across four rounds from November 2020 to June 2023.

The super app firm managers were selected based on the following criteria: (1) a minimum of 3 years of hands-on experience within super app firms, which ensured deep contextual understanding; (2) active involvement in both design and operational aspects of app development; (3) demonstrated expertise in new-age technology through their roles, knowledge, and technical skill sets, and (4) strong experience in identifying and responding to evolving customer needs, particularly in areas like digital payments and mobile wallets. Most participants were based in Asia, where emerging markets drive over 70% of global super app revenues (Statista 2024). Each manager had at least 8 years of industry experience (with a minimum of 3 years specific to a super app firm). This ensures their exposure to strategic processes, customer engagement models, and design-operations integration of super apps.

Further, the sample of *managers from participating firms* was selected based on (1) their transition towards mobile app-based platforms, (2) their future intention to participate as mini apps for super apps if not opted now, and (3) their intention to expand their business presence.

The interview team consisted of one professor and one doctoral scholar. Each telephonic or virtual interview lasted an average of 84 min. In line with theoretical saturation, 70 interviews comprised 14 super app firm managers, 15 participating firm managers, and 41 app customers. Further demographic and affiliation details of the participants are provided in Web Appendix—A4.

The interviews offered a comprehensive view of the super app ecosystem, revealing how design, adoption, and long-term use are shaped by multiple actors. Insights are captured below.

First, customers emphasize the importance of security, trust, and ease of navigation. They appreciated having diverse services on one platform, but also voiced concerns about limited phone storage and information overload. Many preferred super apps that were space-efficient and streamlined, reducing the need for multiple downloads. Social influence also plays a role, as many users adopted apps based on community usage and vendor networks (McPherson et al. 2001). Increasingly, customers expect AI-driven personalization and a smooth, intuitive interface.

Subsequently, super app firm managers focus on expanding features, integrating services, and forming strategic partnerships

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TABLE 2 | Select marketplace evidence for AI-embedded super apps and associated insights.

Super app (market)	Super app positioning and characteristics	Factors for super app development	Evidence
KakaoTalk by Kakao (South Korea)	Optimized app features	– Technology and infrastructure – Strong user base – Contextual Knowledge	KakaoTalk, a Korea-based super app, has integrated generative AI to launch features such as summarizing unread messages and changing the tone of messages in chat rooms based on a lightweight language model (Byung-yeul 2024).
PayTM (India)	Secure digital payment mechanism	– Technology and infrastructure – Privacy and security – Skilled expertise	PayTM, an India-based financial super app, introduced Paytm Intelligence (Pi), leveraging AI and ML to flag possible risks and frauds. It connects data signals from across the customer journey for digital safety and security (Soni 2021).
WeChat by Tencent (China, Europe, US)	Serve diverse services of daily life, providing customized services	– Established customer base – Local market context – Evolving consumption pattern – Partnerships and mergers	WeChat, a China-based super app, uses generative AI to improve video game content and for social media ad recommendations. Tencent built an AI centre in Shenzhen to "Make AI Everywhere," concentrating on research in ML, speech recognition, NLP, and computer vision to build practical AI applications for business in content, online gaming, social, and cloud services (Cheng et al. 2020; Huang and Miao 2021)
GoJek (Indonesia, Vietnam)	Leverage technology and partnership for umbrella services	– Customer traction – Partnership and mergers – Enhance market reach	GoJek, an Indonesia-based super app, has partnered with TenMax.io to provide AI techniques for one-stop advertising solutions to plan, measure, and optimize digital marketing efforts across social media and content sites (Marketing-interactive 2021).
Alibaba by Taobao & Tmall (China)	Combine features with mediation services	– Technology and infrastructure – Skilled expertise – Gain a competitive edge	Alibaba, a China-based super app, provides ML and cloud-based solutions for developers with a rich ecosystem, including components for app infrastructure and business features, containers for running mini-apps, and services for DevOps and operations (Aimagazine 2023)
Avo by Nedbank (Africa)	Digital payment mechanism with diverse services	– Customer base – Security and privacy – Evolving customer needs	Avo, an AI-embedded super app by Nedbank, offers a comprehensive platform that tailors services to customers' lifestyles. AI ensures bank-grade security and facilitates secure payments for enhanced user experience (Tech Central 2020).

to increase engagement and revenue. They highlighted the need for opt-in data-sharing policies that build customer trust while ensuring regulatory compliance.

Simultaneously, participating firms view super apps as high-footfall digital marketplaces. These platforms offer greater brand exposure, wider customer reach, and improved customer loyalty. Firms leverage super apps' AI capabilities and skilled workforce to boost visibility and streamline customer targeting. Although they share profits with super app platforms, firms report a positive return on investment driven by higher sales and market expansion.

Interestingly, super apps and participating firms recognize that emerging markets like India, China, and Indonesia are pivotal for growth. These markets benefit from affordable smartphones, high digital engagement, and fewer regulatory constraints. Success in these markets depends on adapting to local behaviors and cultural nuances, though firms must remain vigilant about privacy and security regulations. Table 3 provides selected interview insights.

In summary, while customers and participating firm managers value super apps, their motivations differ. Customers prioritize

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TABLE 3 | Select stakeholders interviews verbatim and insights.

Data source	Key insights from interviews	Driving forces	Factors and dimensions	Select supporting verbatims	Value creation
Customer	• Lack of phone storage and information overload drives the preference for super apps. • Customers appreciate the convenience, trust, and time saving aspect and of super app, likening them to a digital supermarket. • Technological acceptance is high, with users eager for new offers and new AI features. • Security and reliability of digital transactions are key factors in adopting super apps, replacing traditional methods. • There is a network of dependencies within the community, where the local vendors use the app, influencing users to stay socially connected and relevant. • An inclusive environment, where diverse services and ease of use contributes to digital empowerment, improved lifestyle and networking for customers.	• Limited phone storage • Information load • Need to fit in society • Unreliable payment operations online • Need for social connection	• Technological acceptance • Diverse services • Trust and transparency • AI technology • Network co-dependence • Digital transaction • Inclusive environment • Secured access	• *"Nothing is more promising than the promise of good customer experience with transparency." (M, 34 years)* • *"As time becomes a currency, I will value this app more as I get all things at one place like the supermarket." (F, 42 years)* • *"Space is one thing that people may be concerned about, with so many apps and small retail apps blooming. It is hard to install all the apps on our mobile. Not all android phones have high storage capacity." (M, 26 years)* • *"Super app is my virtual companion, earlier I used to spend much time in the queue to pay for utility bills as online sites were not reliable, but we can bank upon the current super app as the trusted payment will run smoothly." (M, 38 years)* • *"I need a new exciting offer whenever I open any retail app, and we need something that improves our way of living. I look at things online because there are many things, especially new gadgets with advanced evolving technology like AI and virtual reality." (F, 29 years)* • *"Everybody, including local vendors, are using the app. I have to install it to keep up in the society." (M, 24 years)* • *"Being able to do so many things on this one app makes life so much easier. I can chat, shop, and pay bills without switching between different apps." (M, 48 years)* • *"I feel over exhausted with so much information coming from different phone apps like social media" (F, 36 years)* • *"The diverse services available on this app really uplift my way of living. It's like having a digital assistant that helps me manage everything efficiently." (F, 42 years)* • *"With so many apps out there, it's overwhelming, but this super app simplifies everything by bringing all the necessary services together in one place." (F, 36 years)* • *"This one super app helps me fit in socially because everyone else is using it too. I feel like I'm part of a larger community."(M, 24 years)*	• Digital empowerment • Easy to navigate • Improved lifestyle • Social acceptance • Better standard of life • Higher technological adoption • Better customer experience

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TABLE 3 | (Continued)

Data source	Key insights from interviews	Driving forces	Factors and dimensions	Select supporting verbatims	Value creation
Super app firm managers	• Collaborative partnerships enhance the value proposition of super apps by offering diverse services under one platform. • Generative AI is essential for managing and personalizing services, ensuring a cohesive user experience. • Permissions for data sharing are crucial to maintain user trust and compliance with privacy-related government regulations. • Optimized and integrated features within a super app attract and retain customers through engaging AI-driven experiences. • High-end customer base and traction with AI-embedded features backed by firm financial strength and brand directly contribute to improved profits and ROI. • Inclusive business practices and continuous feature updates help maintain customer loyalty and app sustainability.	• Need for consolidation • Limitations of the traditional apps • Evolution of AI technology • Competitive forces of market • Differentiate from standalone apps	• Mergers and partnerships • AI algorithms • Diverse services • Secure access • Government regulations • Optimized features • Financial transaction • Brand image • Financial strength • Firm agility • Technology and infrastructure • Emerging versus developed country • Local knowledge of markets	• *"In India, we need more revenue as the business is of volume and not value anymore. The more features we include, the more customers will stay with us." (M, 41 years)* • *A super app consolidates multiple services within a single platform, requiring generative AI to manage and personalize these services efficiently for users. (M, 38 years)* • *"With multiple features in one super app, we can plan to prioritize products according to the customer's needs & push new features to keep them hooked." (M, 39 years)* • *Without branded mergers and acquisitions, we cannot roll the marketing campaigns within the app." (M, 41 years)* • *"The mobile apps cannot work on their core, now every platform, including social platforms, has added e-commerce, delivery, or payment options." (M, 32 years)* • *"Unlike standalone apps, super app's reliance on AI is fundamental to its operation, driving its ability for cohesive user experience across services." (M, 46 years)* • *"The customers need to be attracted with exclusively optimized features in super app...retained with engaging and interactive experiences with AI." (M, 47 years)* • *"Mostly people prefer a super app to manage their finances that can easily track for utility bills." (M, 43 years)* • *"We have to provide an option to the customer for sharing data, considering government regulations." (F, 35 years)* • *In emerging markets like Mexico or Brazil, the expectations of the customers are evolving." (M, 38 years)* • *"With our financial backing and strong brand reputation, we can push AI-embedded newer features to keep customers hooked on to the app. Our business practices need to be inclusive for sustainability." (M, 39 years)* • *"Having business through apps is essential for businesses. It is hard for them to sustain one app as it requires a workforce. Super app have cashed this problem of small businesses to achieve long term engagement." (M, 46 years)* • *"Customers in India love clutter and might prefer a super app whereas super apps are not a popular concept in the US and Europe." (M, 41 years)*	• Increased margins • Customer opt-in • Increase customer base • User experience • Customer traction with AI • Improved revenue and profits • Inclusive business • Sustainable app

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TABLE 3 | (Continued)

Data source	Key insights from interviews	Driving forces	Factors and dimensions	Select supporting verbatims	Value creation
Participating firms	• Exposure in super apps provides unparalleled reach, akin to high-traffic retail environments, boosting customer retention. • Leveraging advanced technical support and a skilled workforce from super apps enables smaller firms to succeed. • Profit sharing with super apps yields substantial returns due to higher inquiry rates and sales conversions. • Localized understanding is key to effectively addressing cultural nuances and societal behaviors in different markets. • Access to extensive customer bases and the ability to reach diverse audiences expand market presence and brand influence. • Strict adherence to data privacy regulations is necessary to mitigate concerns over sharing customer information. • Enhanced customer engagement and loyalty stem from the convenience and visibility afforded by the super app framework.	• Advanced features • Need for convenience • Data privacy concerns • Limited standalone capabilities • Competitive edge in the market	• Hyperlocal knowledge of markets • Technology expertise • Technical Support • Skilled workforce • High customer base • Government rules and regulations • Customer loyalty • Cultural and social aspects of emerging markets	• *"My product is visible to a range of people, which I have not dreamed of in this short period." (M, 42 years)* • *"Super app is like being present in high footfall mall, we get traffic and loyal customers." (F, 38 years)* • *"Big giants launching super apps have the money, AI technical support, and skilled employees." (M, 45 years)* • *"Although I need to share profit percentage with the platform, it still gives better ROI regarding greater enquiries and better conversion rate for the sales; we enjoy massive market share with better experience." (M, 44 years)* • *"The markets need to be studied for the cultural aspects, societal expectations and behavior." (M, 39 years)* • *"Sharing the entire customer data with the platform firm is a concern; strict regulation needed." (M, 42 years)* • *"We have an edge in India where the government is not that restrictive like the US and other developed countries regarding market entry, technological imports, or even data access for personalization." (M, 42 years)* • *"Customers download our mini app version through the super app, and we can widen the market base." (F, 39 years)* • *"Smart products like these can do conversations with each other, and as a fact a person spends more time on their gadget than any other living person. The connection of everyday chores to the app will make picking the concept of super app really quickly in any market." (M, 38 years)* • *"So super app as a concept, it is going to be difficult to implement but everyone is going for it because the margins in any one particular sector are very minimal." (M, 43 years)* • *"The super app requires huge capital to become a distinctive player in India, even after partnerships and mergers. It has to be built by a big brand." (F, 39 years)* • *"Affordable data plans disrupt the market and help app companies grow. So big telecom operators have a major role to play in future to increase customer acquisition, with the introduction of 5G and cheaper data plans...it's going to help the bottom of the pyramid to accept digital technology." (M, 42 years)*	• Market visibility • Influential brand image • Improved ROI • Higher market share • Stickiness of customers • Customer loyalty and traffic • Better customer experience

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convenience, multifunctionality, and trust, whereas firms focus on market visibility, ROI, and platform scalability. Super app providers must balance these differing expectations, offer user-centric design while sustaining growth and stakeholder value.

All interviews and feedback were recorded for verbatim content analysis. The interview process was iterative and continued over the years following the Gioia method (Gioia et al. 2013), allowing for conceptual development. Next, we apply thematic analysis with triangulated insights from three key sources—literature review, marketplace evidence, and interviews.

3 | Thematic Analysis and Coding

The triangulated insights from the literature review, marketplace evidence, and interviews provided us with academic and industry perspectives on super app adoption and engagement. Further, we conducted a thematic analysis of these triangulated insights. Thematic analysis systematically codes and categorizes textual data to identify patterns, trends, and relationships (Vaismoradi et al. 2013). We conducted thematic content analysis for two reasons: (1) to identify the key attributes and underlying dimensions of super apps, and (2) to develop a process framework for their design, adoption, and sustained engagement.

A panel of eight reviewers was assembled for this task, comprising three professors (from China, India, and the USA), two PhD researchers (from India and the UK), two industry experts (from the UK and India), and one research assistant (India). The panel independently coded the qualitative data and collaboratively identified recurring themes and categories. These were then organized into first-order (descriptive) and second-order (conceptual) factors.

To ensure reliability, a subset of a four-member panel, comprising two PhD students and two research assistants, evaluated the extracted factors and dimensions to validate the coding structure. Agreement was rated using a 5-point Likert scale, and an inter-coder reliability test was conducted using Fleiss' Kappa. After applying the negotiated agreement method, Fleiss' Kappa reached 0.84, indicating strong consensus and reliability in coding. For face validity, the resulting themes and dimensions were shared with two super app firm managers in the final analysis phase. Their feedback helped refine several categories, ensuring the relevance and practical validity of the findings. Table 4 provides the details of the coding process and structure.

In summary, the thematic analysis established a reliable and valid coding structure that captures the multidimensional nature of the super app ecosystem. Building on this foundation, we now define the super app construct, explore its core dimensions, and identify the key drivers of adoption and engagement that contribute to long-term value creation.

4 | Defining Super App–Construct and Associated Dimensions

There has been a lack of an inclusive definition explaining the unique characteristics of the super app (Nayebi and Ruhe 2017; Steinberg 2020). Employing the thematic analysis process and coding, we identify the critical dimensions defining the super apps (see Table 1: Section 2 and Table 4). We also compare super apps and standard mobile apps across various dimensions (Web Appendix—A1). Based on the conceptualization, we define a super app as "a multi-purpose mobile application that serves as a mediation platform for diverse utility functions—enhanced by generative artificial intelligence—deployed with optimized features within the individual app environment and integrated with a digital payment mechanism." Next, we discuss all the dimensions in detail.

Integrated digital payment mechanism allows users to perform various financial activities, such as paying bills, transferring money, and making purchases, within a unified platform, without switching between individual mini-apps (Roa et al. 2021). One interviewee noted, "Mostly people prefer a super app to manage their finances that can easily track utility bills" (M, 43 years). Generative AI enhances payment systems by enabling secure transactions and personalized experiences. It analyzes user behavior, preferences, and past transactions to offer intelligent payment routing, fraud detection, and natural language-based search functionality.

Mediation platform positions the super app as a connector between participating firms and customers (Steinberg 2020), consolidating services in a single space and allowing users to interact with multiple providers without leaving the platform. Integrating generative AI, the platform personalizes and automates key business functions like advertising, marketing, and operations, enhancing firm efficiency and customer engagement (Kumar and Gupta 2016).

Inclusivity ensures that diverse stakeholders can access, interpret, and benefit from the platform seamlessly. Generative AI facilitates this by supporting local language use, personalized recommendations, and adaptive accessibility features. As one interviewee highlighted, "With our financial backing and strong brand reputation, we can push AI-embedded newer features to keep customers hooked on to the app... need to be inclusive for sustainability." (M, 39 years). This dimension also opens opportunities for small and medium enterprises to participate meaningfully within the super app ecosystem and access tailored products and services for every customer segment (Zhu et al. 2023).

Individual app environment refers to the distinct and self-contained interface of mini-apps, offering a distinct experience while remaining connected to the larger ecosystem (Cheng et al. 2020). Generative AI is critical in managing interoperability and user experience, allowing smooth transitions between mini apps while retaining their unique brand identities (Glikson and Woolley 2020). This ensures consistency without compromising individuality.

A multi-purpose single platform is a digital space combining various features, services, and user experiences into one cohesive interface (Gokgoz et al. 2021). AI integration supports this consolidation by streamlining navigation, filtering irrelevant content, and recommending relevant services based on user needs. Unlike traditional apps that serve a single purpose, super apps offer sector-spanning utility, making them indispensable for everyday digital activity (Web Appendix—A5).

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TABLE 4 | Thematic analysis coding and categorization of framework components.

Zero order coding	First order	Second order	Third order
– Owns an independent payment system and encourages users to make transactions (Santana 2019; O'Brien 2022). – Providing end-to-end transaction security and hassle-free processing through generative AI-enabled transactions.—Perform various financial activities like paying bills, transferring money, and making purchases without switching between different applications.	Integrated digital payment mechanism	Super app dimensions	
– Connecting service providers and customers via financial transaction platform (Steinberg 2020). – Centralized access to participating and third-party apps using AI functions.—Centralized facilitation of marketing activities for participating firms.	Mediation platform		
– Strengthen digital inclusion of multiple lightweight apps for service providing participating firms. – Ensured easy access for all segment customers transiting to digital app with generative AI-enabled interaction.—Encourages social responsibility through aggregate services with environmental impact (Sherlock 2021).	Inclusivity		
– Allows participating firms to enter the platform within its exclusively intact ecosystem (Cheng et al. 2020). – Customers have an easy transition from the platform app to the individual app of participating firms using AI.—Provides an authentic app-like experience for each participating firm to the customers.	Individual app environment		
– Offering a range of services for conducting a variety of daily tasks under one platform powered by generative AI. – Bundles of stimuli, functionalities and experiences (Gokgoz et al. 2021).—Platform's ability to replace multiple standalone apps by combining essential services into one platform.	Multi-purpose single platform		
– Tool for super app usability with efficient and effective use of lightweight micro-apps (Cheng et al. 2020). – Cohesiveness between newly added and existing features (Nayebi and Ruhe 2017).—Optimizes ROI of participating firms in terms of higher perceived value (Zeuxinnovation 2020).	Optimized functionality		
– Digital customer orientation with altering preferences for multiple products and services (Stocchi et al. 2022). – Customers are becoming aware of apps that can satisfy their altering needs and convenience (Gokgoz et al. 2021). – Increasing use of mobile phones with distinct consumption and touchpoints (Stocchi et al. 2022).	Evolving consumption patterns	Customer-related desirability	Desirability
– It is convenient for users to identify and find the required features, products, or services. – Availability of relevant information and offerings in one place in an organized way (Kumar et al. 2019). – Frictionless and smooth interaction with the highly defined user interface of the app (Stocchi et al. 2022).	Ease of navigation		
– Customer has confidence and certainty in personal data security (Gokgoz et al. 2021). – The app holds a positive perception with reliable associations, connections, and information.	Secure access/ trust		

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TABLE 4 | (Continued)

Zero order coding	First order	Second order	Third order
– User is invested in app-based products and transactions frequently (Kumar et al. 2019). – High reviews and ratings for the app with a continuous usage intention. – Paying or non-paying customers subscribed to the product or service.	High level customer traction	Firm-related desirability	
– Ability to respond to competitive moves, differentiated offerings, unique market positioning (Kumar et al. 2019). – Focus on providing superior customer experiences than other firms.	Competitive edge		
– High downloads and transactions through the app for market usability (Gokgoz et al. 2021). – A higher estimated number of potential customers that the super app can reach. – Profitable retention of users with large impressions relative to the frequency of exposure to app.	Enhanced market reach		
– User's propensity to embrace and use cutting-edge technologies (Blut and Wang 2020). – Interested in technology-mediated services via smartphones with processing power. – Positive attitude and openness to adopting new things, experimentation, and exploration.	Technology readiness	Customer related viability	Viability
– Adaptive hardware for the app within the user's reach, that is, a phone equipped with the required storage. – Users can own smartphones with sufficient storage and operating system compatibility to run the app smoothly.	Hardware accessibility		
– The firm has a customer base from other businesses with a high customer retention rate (Kumar et al. 2019). – The firm owns a significant market share from other businesses (Kumar et al. 2019). – Able to generate customer traffic (Gokgoz et al. 2021).	Strong user base	Firm related viability	
– The business supports through all phases of economic cycle with stable revenue, owns liquid and capital funds. – Platform firm showcases high shareholder value, sales, profits, and assets.	Financial strength		
– Pioneer image in adopting new features for a holistic brand experience (Gokgoz et al. 2021). – Market recognition of the established brand associated with the app (Kumar et al. 2019). – Intention of stakeholders to be associated with the firm powering the app (Stocchi et al. 2022).	Affirmative brand image		
– Customization at a granular market level adapted for relevant content (Gokgoz et al. 2021). – Location sensitivity in decision-making for relevant content (Gokgoz et al. 2021). – Adapt to the changing market landscape with actions at the customer level (Kumar et al. 2019).	Contextual marketplace knowledge		
– Ability to lead change, adapt in uncertainty and seize opportunities (Arteta and Giachetti 2004). – Efficiency in operational performance and cooperation to enhance competitiveness in responsive strategies. – Redirection of people and priorities towards value-creating actions (McKinsey 2022).	Firm agility		

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TABLE 4 | (Continued)

Zero order coding	First order	Second order	Third order
– Availability of shared information technology resources of hardware and software. – Availability of a real-time data compilation system in place. – High-speed internet access via mobile networks at a low price.	Technology and infrastructure	Macro factors	Feasibility
– Barriers to entry into the mobile app industry with strict policies for access, handling, and protection of data. – Conducive business regulation for web-based allied partners with security practices and procedures.	Government regulations		
– Effective buyer–supplier relationship and communication network. – Collaborations and associations with other participating firms in data exchange and sharing. – Integration with the partner firms to co-produce an innovation.	Partnerships and mergers	Macro factors	
– Availability of experts with the ability to generate required human capability when needed. – Effective human resource initiatives for technological and technical advancements. – Engaged employees to manage ever-changing IT and software issues.	Skilled expertise		
– Leverage signaling and information to access benefits among the network groups. – Customers embedded in key network ties for mutual and balanced interdependence. – Visibility among the structured and social ties with word-of-mouth volume (Kumar et al. 2019).	Network co-dependence	Moderators	Moderators
– Infrastructure (physical, financial, technological, and social) (Kumar et al. 2019). – Cultural norms and values of the users. – Economic growth patterns and market potential (Kumar et al. 2019).	Emerging versus developed country		

Optimized functionality refers to the seamless integration of new and existing mini apps, supported by back-end systems that maintain system stability and performance (Nayebi and Ruhe 2017). Generative AI enables faster adaptation by managing compatibility, supporting updates, and ensuring minimal disruption as services are added. Users benefit from a sense of continuity and reduced friction, perceiving greater value as the platform evolves with minimal onboarding effort (Zeuxinnovation 2020).

In conclusion, these six dimensions, enhanced through generative AI, are foundational to super app success. The depth and balance with which they are implemented shape user experience, firm participation, and long-term platform sustainability. In the following section, we present a process framework for super app development and propose key research propositions.

5 | A Process Framework for Super App Design, Customer Engagement and Value Creation

Understanding the key factors that drive the development, adoption, and sustained engagement of super apps is essential for unlocking their full potential across diverse markets. To address this, we propose a process framework for super apps grounded in the thematic analysis of triangulated insights and guided by a human-centered design thinking approach (Figure 1). We adopt a design-based approach, where the super app is developed iteratively by embedding AI into the user interface. Customer interaction data is treated as feedback to refine flows and features, constituting a dynamic, data-informed approach to digital engagement. The framework identifies (1) the key forces driving the need for super apps; (2) the role of design elements in shaping user behavior; and (3) how context and engagement dynamics influence long-term value creation. Additionally, it integrates customer engagement theory to explain how enhanced utility drives sustained user interaction and business value. Given the limited academic focus on AI-driven super apps, our framework offers a structured lens informed by qualitative insights and existing theory.

The first layer outlines the motivating forces across stakeholder groups, customers, super app firms, and participating businesses. These include demands for seamless digital experiences, payment integration, personalization, and market visibility.

The second layer draws on the principles of design thinking, a problem-solving approach that prioritizes user immersion, rapid iteration, and feedback-driven improvement (Chasanidou et al. 2015). Unlike traditional, linear product development, design thinking operates through three fluid, interconnected elements—viability, desirability, and feasibility—allowing

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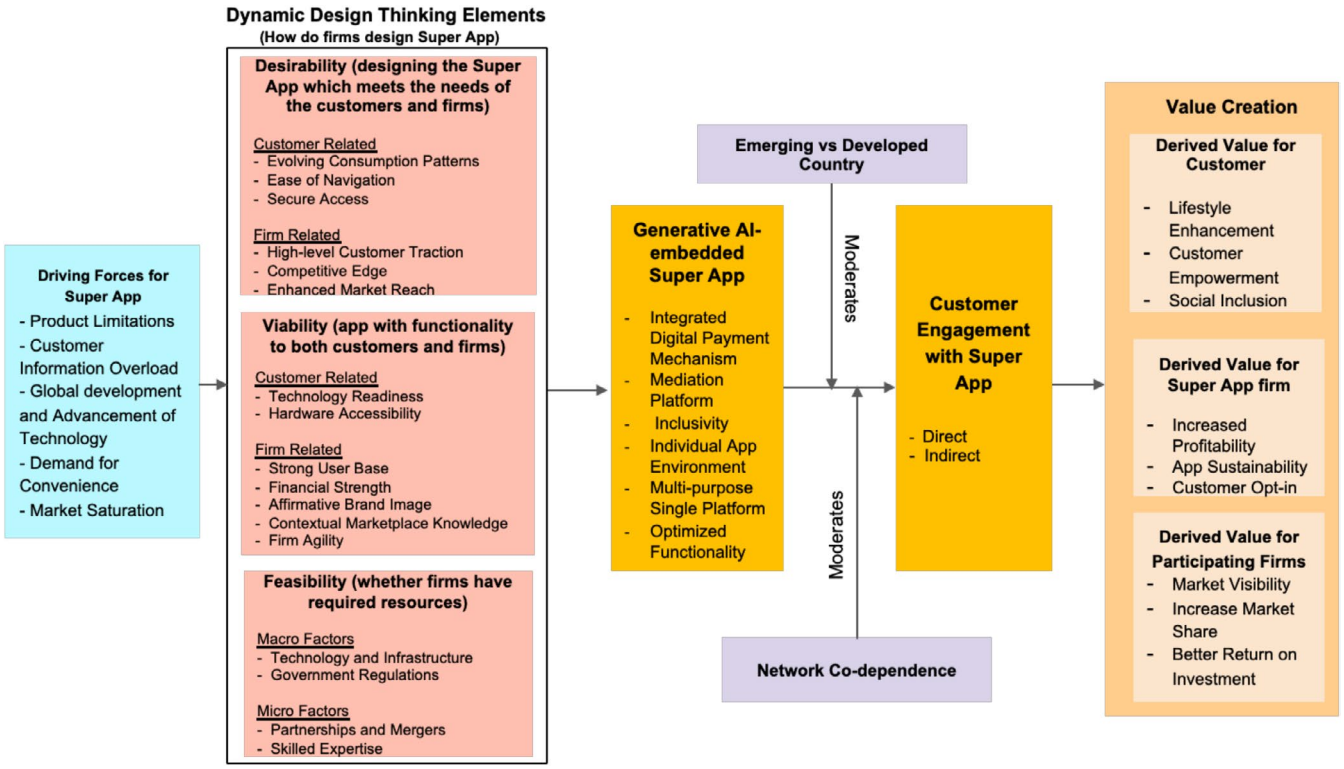


FIGURE 1 | Dynamic design thinking generative AI-embedded super app: A process framework towards customer engagement and value creation.

innovation to emerge dynamically rather than following a rigid step-by-step process (Brown 2008; Chasanidou et al. 2015). The desirability aspect focuses on how users interact with and perceive the super app's functionality. The feasibility lens assesses whether technological and organizational capabilities exist to support development. Viability ensures the app's business model is economically sustainable and aligned with long-term market success. These elements work together to ensure the super app is useful, buildable, and profitable, adapting dynamically to market feedback.

The third layer integrates customer engagement theory (Pansari and Kumar 2017), showing how improved utility, trust, and customization deepen user participation. Customer engagement occurs at two levels: direct engagement, through continued app usage and feature adoption, and indirect engagement, through word-of-mouth recommendations, referrals, and user-generated content (Kumar et al. 2019). Six AI-enabled design dimensions of a super app are enablers of this engagement, driving a loop of increased utility and customer retention. The framework also highlights the moderating role of market context. Adoption and engagement patterns vary between developed and emerging markets, influenced by infrastructure maturity, regulatory conditions, and cultural norms. For instance, emerging markets may prioritize mobile-first access and payment flexibility, while developed markets may emphasize data privacy and app modularity.

In sum, the framework provides a systematic approach to understanding how design, engagement, and context interact in shaping the success of super apps. It offers a roadmap for firms to design platforms that attract users and build long-term value through meaningful digital experiences. The following section explores these components in greater depth and outlines key research propositions emerging from this framework.

5.1 | Driving Forces for Super App

The rise of super apps is shaped by a convergence of market, technological, and user-centric forces, creating a strong case for integrated digital platforms. These drivers reflect firms' efforts to overcome product limitations, manage information overload, meet consumer demands for convenience, technological advancements, and remain competitive in saturated markets.

The first key driver is *product limitations*, particularly in emerging markets where low-cost Android devices with limited storage and processing capacity restrict users from downloading multiple standalone apps (Mekuria and Mfupe 2017; Watson and Sauter 2011). Customer interviews highlight this concern, with one user noting, "Space is one thing that people may be concerned about, with so many apps and small retail apps blooming. It is hard to install all the apps on our mobile." (M, 26 years). Super apps address this issue by consolidating multiple services, enhancing accessibility while reducing storage constraints.

Second, *customer information overload* arises when users struggle with managing excessive notifications and scattered digital interactions across multiple platforms. Many users expressed frustration with content fragmentation, with one stating, "I feel over-exhausted with so much information coming from different phone apps like social media" (F, 36 years). Super apps help centralize communication and services, allowing users to

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manage their digital interactions more efficiently and avoid cognitive fatigue (Kumar et al. 2019).

Third, the *demand for convenience* reflects society's growing emphasis on comfort and ease in daily transactions. Users now expect seamless digital experiences integrating communication, shopping, banking, and entertainment into one platform (KPMG 2019). One respondent said, "*Super apps feel like a digital supermarket. As time becomes a currency, I will value this app more as I get all things in one place" (F, 42 years)*. By offering multi-functional accessibility, super apps reduce friction in daily transactions and enhance user satisfaction.

Fourth, the *global development and technological advancement* have accelerated super app expansion, particularly in generative AI-driven platforms, mobile payments, and smart recommendations to personalize services and enhance security. For instance, PayTM introduced AI-powered fraud detection systems to ensure safer transactions (PayTM, 2022), while WeChat uses generative AI-driven recommendations to optimize user engagement (Marr 2018).

Fifth, *market saturation* has pushed firms toward multi-functional platforms, as standalone apps struggle to retain user engagement. As one manager explained: "In India, we need more revenue as the business is of volume and not value anymore. The more features we include, the more customers will stay with us." (M, 41 years). This underscores the strategic advantage of super apps, which sustain engagement by offering an integrated ecosystem (Chasanidou et al. 2015).

Together, these forces create a compelling case for super apps, shaping the digital landscape by responding to consumer needs and leveraging AI-driven solutions. Thus, we propose:

P1. Market forces such as (a) product limitations, (b) customer information overload, (c) demand for convenience, (d) global development and advancement of technology, and (e) market saturation, drive the need to develop super apps in any given context.

As these forces continue to shape digital business models, the following section explores the critical design dimensions essential for long-term value creation and engagement.

5.2 | Design Elements for Super App

Insights from thematic analysis reveal that human-centered design thinking is crucial for developing super apps, incorporating three core elements: desirability, viability, and feasibility (Brown 2008; Chasanidou et al. 2015). These elements collectively ensure a user-centric, scalable, and sustainable platform for enhanced adoption and engagement (see Figure 1; Table 4).

Desirability within the design thinking framework focuses on meeting user expectations and business objectives, leveraging innovations to enhance digital interactions. (Brown 2008; Chasanidou et al. 2015). Based on the thematic analysis, we identify customer-and firm-related factors crucial for desirability elements essential for developing a successful super app.

Customer-related factors such as changing consumption patterns, ease of navigation, and secure access to services reflect the customer-based justification for having a super app.

Evolving consumption patterns reflect a growing digital orientation among customers (Stocchi et al. 2022). Marketplace evidence supports this trend: LINE (Japan, Taiwan, US) expanded beyond messaging to offer e-commerce, social media, and financial services, improving customer retention and engagement (Steinberg 2020).

Ease of navigation underpins the convenience for users to locate the desired services easily (Stocchi et al. 2022). This sense of convenience enhances the overall user experience, promotes repeated use, and strengthens engagement with the super app (KPMG 2019).

Secure access ensures that the platform is secure with encryption and firewall protection and that user data is safe from any misuse or data hacking. For instance, WeChat (China) employs AI-driven recommendations to streamline daily transactions (Marr 2018). These super apps mitigate privacy-related threats for stakeholders and assure users that any third party will not share their personal information or financial data without their consent (Gokgoz et al. 2021).

Firm-related factors such as high customer traction, competitive advantage, and enhanced market reach contribute to the desirability element of super apps driving their development.

High customer traction involves the number of paying and non-paying users who frequently download and use the app (Kumar et al. 2019). It signals strong user investment, where customers frequently rely on the app for transactions and interactions. High traction is also reflected in positive reviews, high ratings, and continued usage intent (Gokgoz et al. 2021).

Gaining a *competitive edge* by outperforming its rivals in the market is crucial for a firm to take the lead in creating a super app. A firm can secure substantial competitive advantages by staying updated with emerging technological trends, such as generative AI for innovative solutions (Kumar et al. 2019). Businesses that bundle services on a single platform increase engagement, as evident in Careem's (Dubai) AI-driven security features, personalizing user interactions while ensuring compliance with data privacy regulations (McKinsey 2022).

Enhanced market reach signifies increased downloads across customer segments, higher app traffic, and lead generation through the app (Lee 2018). Interview insights highlight this focus: "We need more revenue as the business is of volume and not value anymore. The more features we include, the more customers will stay with us." (M, 41 years). This underscores the importance of bundling services and market positioning to maximize long-term engagement.

Viability encompasses the critical factors required to ensure a product's functionality and sustainability (Seidel and Fixson 2013; Hehn and Uebernickel 2018). This element of the design approach outlines and quantifies the requirements for

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designing a super app based on customer and firm-related factors that highlight the crucial viability elements.

Customer-related factors highlight the significance of customers' technology readiness and ease of hardware accessibility, which determine super apps' scalability.

Technology readiness refers to the extent of customers' adoption of technology and their willingness to embrace advanced technological solutions (Blut and Wang 2020; Gupta et al. 2020; Kumar and Pansari 2016). Technology-ready customers are inclined to adopt super apps to address their daily needs more experientially and satisfyingly, enhancing their engagement.

Hardware accessibility refers to customers' ability to access smartphones with the processing power needed to handle complex AI-embedded super apps (Gokgoz et al. 2021). The growing availability of affordable, high-functioning smartphones has enabled broader user participation in digital platforms (Watson and Sauter 2011). This is highlighted in the interviews: "Affordable data plans disrupt the market and help app companies grow... with 5G and cheaper data plans making digital technology accessible to more people." (M, 42 years).

Firm-related factors under the viability element suggest that a firm should have a substantial user base, financial robustness, an affirmative brand image, contextual marketplace knowledge, and agility to influence successful app adoption and user engagement.

A *strong user base* demotes an established network of active customers across a firm's business segments, creating synergies that drive sustained traffic, facilitate new product adoption, and support monthly active user retention (Web Appendix—A5). Over time, this leads to greater user stickiness and market share (Gokgoz et al. 2021; Kumar et al. 2019). Additionally, the effectiveness of generative AI depends on access to robust user data, enabling personalized recommendations through insights into user preferences.

The firm's *financial strength* reflected in capital reserves, shareholder value, and stable revenue streams plays a vital role in super app development. As one firm manager noted, "With our financial strength and strong brand reputation, we can push AI-embedded features to keep customers hooked." (M, 39 years). Financially stable firms are better positioned to invest in design, innovation, and marketing, thereby increasing the likelihood of successful super app adoption and sustained customer engagement (Web Appendix—A5).

The *affirmative brand image* explains the firm's positive influence and reputation, shaped by customer attitudes, perceptions, and beliefs (Kumar et al. 2019). This affirmative image creates a halo effect, encouraging users to explore new features, adopt bundled services, and remain engaged within the app's ecosystem. Brand equity is a key driver of user acquisition and engagement for super apps competing in crowded markets.

Contextual marketplace knowledge refers to a firm's ability to comprehend marketplace dynamics, including its patterns and disruptions (Gokgoz et al. 2021). Several mobile app firms recognize the need to study the markets "for the cultural aspects, societal expectations and behavior." (M, 39 years). Firms with such knowledge are better prepared to indulge in developing super apps while addressing existing contextual needs, driving traction and usage (Zhu et al. 2023).

Firm agility is an organization's ability to quickly sense, adapt, and respond to external changes to maintain a competitive edge (Ghasemaghaei et al. 2017). Super apps developed within agile firms that embed generative AI across operations are better positioned to make informed decisions and drive sustainable innovation.

Thus, the interplay of all the above factors collectively contributes to a firm's potential for successful app development and user retention.

Feasibility ensures that the firm has essential resources, a supporting environment, crucial infrastructure, and key partners available for developing and designing products or innovation (Brown 2008; Chasanidou et al. 2015). The feasibility element considers 'whether the firm has required supporting resources' and 'what is technically and organizationally achievable' (Hehn and Uebernickel 2018) to translate the human-centered requirement into actual products, services, or processes. The following *micro* and *macro factors* are critical in determining the feasibility of designing the super app and enhancing customer engagement.

Macro factors, such as technological infrastructure and government regulations, are crucial in successfully introducing super apps in the given context.

Technology and infrastructure form the foundational architecture needed to support the complex design and development of super apps (Varshney 2020). The effective use of advanced communication technologies, including AI and data systems, is essential for creating user-friendly interfaces. Additionally, affordable data plans and reliable mobile internet are crucial for enabling widespread usage. Countries like China (WeChat, Alipay), India (Paytm), and Indonesia (GoJek) illustrate how robust infrastructure and low-cost internet access create favorable conditions for super app adoption (Web Appendix—A5).

Supportive *government regulations* facilitate ease of business operations, low entry barriers, and a conducive business environment. Moreover, policies and regulations encompass legal, license, and tax processes influencing market entry and business investment decisions across countries. As one super app firm manager clarified, "We have to provide an option to the customer for sharing data, considering government regulations." (F, 35 years). Hence, a balance by policymakers can be stimulative for firms with their super app initiative.

The *micro factors* pertain to the firm's immediate internal environment, which may significantly impact the firm's decision-making processes related to the super app. Firms are likely to observe a higher level of app usage and engagement in the favorable presence of certain firm-based micro factors, such as strategic partnerships, mergers, and skilled expertise.

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In the mobile app context, *strategic partnerships and mergers* refer to collaborative relationships and cooperative arrangements between two or more organizations encompassing revenue-sharing agreements, cross-promotion strategies, or co-development efforts (Van Der Vlist and Helmond 2021). Co-producing super apps requires robust integration with partner organizations and other key stakeholders such as technology firms, internet service providers, content creators, communication networks, research and development institutions, and firms participating in mini-app ecosystems. Moreover, collaborative synergy across multiple stakeholders is essential for a cohesive ecosystem, enhancing the user experience.

The presence of *skilled expertise* within the firm involves domain experts, IT and technology experts, user researchers, business specialists, and a project lead (Hehn and Uebernickel 2018), which influences the ease and process of research, design, and firm-level initiative of developing a super app. These micro factors collectively underscore a firm's internal dynamics' crucial role in successful engagement with the super app landscape.

In summary, an AI-powered super app would have great potential if its development process underpinned design thinking principles, that is, desirable, feasible, and viable process elements, making it a noteworthy innovation that customers can embrace widely (Dinner et al. 2015; Seidel and Fixson 2013). The iterative process (feedback loop, see Figure 1) of the human-centric design thinking approach garners innovative solutions for the dynamic digital environment (Chasanidou et al. 2015), compared to the traditional problem-solving approach, which emphasizes linear, function-specific, and problem-isolated solutions (Meinel et al. 2020). The traditional problem-solving approach for app development typically defines a single problem, creates a solution, and releases it without continuous user feedback (Seidel and Fixson 2013; Hehn and Uebernickel 2018). It prioritizes adding features in the app to address an issue rather than understanding broader user needs, resulting in siloed, standalone apps (Brown 2008). Additionally, the traditional problem-solving approach addresses localized problems without scalable, multi-industry integration, whereas the design thinking approach helps super apps expand across multiple verticals integrated into a unified ecosystem (Gokgoz et al. 2021; Meinel et al. 2020).

Customer adoption of such user-centered innovations with higher customer influence, word-of-mouth, referrals, and recommendations, network knowledge sharing, and feedback to the associated firm leads to higher customer engagement (Dinner et al. 2015; Venkatesan 2017). Unlike traditional standalone mobile apps, super apps may boost user retention by ensuring *higher super app usage, word of mouth spread, referral, and feedback to the app firm*. Hence, the suitable balance and interaction among all the factors presented under three elements of the design thinking approach—desirability, viability, and feasibility (Meinel et al. 2020)—are likely to lead to successful super app development, resulting in better app adoption and engagement. Thus, we suggest the following:

P2. The intersection of user-centric design elements- viability, desirability, and feasibility- positively influence the process of developing and adopting a dynamic AI-embedded super app than an app developed based on a traditional problem-solving approach.

5.3 | Customer Engagement in Super App

Customer engagement is the "mechanics of a customer's value addition to the firm either through direct or/and indirect contribution" (Pansari and Kumar 2017). In the context of super apps, this engagement is crucial for driving sustained usage, brand loyalty, and network effects. Hence, customer engagement can be categorized as direct and indirect. Direct customer engagement occurs when customers actively use the app for making purchases, subscribing to services, and engaging in activities that contribute to customer lifetime value (CLV) (Dinner et al. 2015; Kumar and Pansari 2016). Indirect customer engagement encompasses non-transactional but high-impact contributions, including customer participation and involvement with incentivized referrals bringing customer referral value (CRV), social media conversations about the brand leading to customer influence value (CIV), and knowledge through customer feedback and co-creations with the firm bringing customer knowledge value (CKV) (Kumar and Pansari 2016). Research shows that customer engagement has a "direct and positive influence on customer stickiness as well as an indirect influence through value creation" (Zhang et al. 2017). This goal of stickiness aligns closely with the core value proposition of super apps, where users remain actively engaged due to convenience, personalization, and continuous value delivery. Firms design super apps with this in mind, combining AI-embedded systems and human-centered design principles to ensure that functionality, feasibility, and user needs are all aligned. As a result, users are more likely to adopt and continue using apps that adapt to their preferences and provide integrated experiences across services. The integration of generative AI further amplifies engagement by enabling real-time personalization, improved data security, intuitive user interface, and holistic digital experiences (Glikson and Woolley 2020). These features go beyond individual transactions, reinforcing the app's value as an all-in-one digital solution and enhancing both direct and indirect user contributions. Thus, engagement-oriented, design-thinking-driven super apps generate value across multiple dimensions: CLV (through repeat usage), CRV (through referrals and incentives), CIV (via social influence and advocacy), and CKV (through user feedback and product input) (Meinel et al. 2020; Venkatesan 2017) (Table 5).

Further, we explore critical moderating factors that specifically influence the impact of design thinking AI-embedded super app adoption on direct and indirect customer engagement.

5.4 | Linking Generative AI-Embedded Design-Based Super App to Customer Engagement–Moderating Variables

Triangulated insights from the literature, marketplace evidence, and stakeholder interviews suggest two key contextual factors: (1) market context of emerging versus developed countries and (2) network co-dependence moderate the relationship between super app design and customer engagement outcomes.

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TABLE 5 | Direct and indirect customer engagement through super app.

Direct effect	Meaning	Marketplace evidence	Value for customers	Value for firms	Challenges
CLV (customer lifetime value)	CLV measures the profit value of 1 unit of brand value perceived by customers and willingness to be associated in future.	Go-Jek, loyalty Go Points uses real-time metrics to measure CLV and customize promotions and offers accordingly. GoJek contributes $732 million to Indonesia's GDP.	Go Point via participating firms turned into a user-friendly solution. Users get reward points for purchases.	Transaction volume across services creates a virtual persona of customers and helps firms customize communication, leading to better performance.	Ensuring CLV and profits is challenging as super apps operate with high investment and low margins.
Indirect effect	**Meaning**	**Marketplace evidence**	**Value for customers**	**Value for firms**	**Challenges**
CRV (customer referral value)	CRV stands for providing referral and works in exchange for the credits/points	PayTM, an India-based super app, focuses on retention and acquisition round-the-year launch schemes such as '3 pe 300' (i.e., 3 referred users will get the user cashback of Rs 300)	Emerging markets are price sensitive. Referral rewards allow customers to earn revenues by using their peer network.	Lower customer acquisition cost. Referral rewards attract existing customers to use the platform and bring potential ones.	The calculation and data collection for the referrals require an efficient process.
CIV (customer influence value)	CIV measures the monetary value of a customer's word of mouth in exchange for credits/points	Alipay, a China-based super app, builds a closed-knit ecosystem with referrals via transactional data of existing customers.	Provide them with a secure credit profile to make them feel safe to share the information.	The measure helps firms to increase better customer acquisition and profitability.	Any reduction in reward incentives can lead to negative WOM.
CKV (customer knowledge value)	CKV measures the value of feedback/suggestions received from the engaged customer.	Rappi, a Colombia-based super app, analyses customer journey data and feedback knowledge to improve the app and customer engagement.	Big data helps the platform to provide relevant recommendations to customers.	Existing customers' purchase patterns and feedback help brands modify/stay-put strategy.	Strict government regulations for data could restrict the platform and affect the revenue model.

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Emerging versus *developed countries*, as moderators, dictate the market's contextual significance to understand various country-related factors such as demographics, culture, resources, government regulations, and business norms (Shankar and Narang 2020). According to the insights from the triangulation approach, emerging markets have shown a trend for successful super app engagement, such as WeChat in China, KakaoTalk in Korea, and GoJek in Indonesia. The exhibition of such success in emerging markets compared to developed markets can be attributed to the following interrelated drivers. First, the technological leapfrogging of smartphones, particularly low-cost Android devices with widespread internet access, has accelerated mobile-first digital adoption in these markets (Mekuria and Mfupe 2017; Watson and Sauter 2011). This has increased direct customer engagement, as mobile apps become the default gateway for digital services. Second, government regulations, such as policies regarding data privacy and information sharing in emerging markets, are considered favorable by the super app firms (Luo 2003). One of the participating firm executives also highlighted during the interviews that, "We have an edge in India where the government is not that restrictive like the US and other developed countries regarding market entry, technological imports, or even data access for personalization." (M, 42 years). Such conditions enable firms to integrate AI-based features with greater ease, enhancing both user experience and business scalability. Third, the consumer's path to purchase and customer journeys differ across countries and cultures (Luo 2003). This is further reiterated as one interviewee pointed out, "In emerging markets like Mexico or Brazil, the expectations of the customers are evolving." (M, 38 years). With the rise of community-based consumption due to collectivist culture, shared values, and aspirational customers in emerging markets (Markus 1987), the super app's adoption in emerging markets positively showcases higher indirect customer engagement through word-of-mouth, peer influence, and referrals (Zhu et al. 2023). Fourth, the widespread use of UPIs and mobile wallets in markets like India and Indonesia supports seamless transactions and lowers the friction of using a consolidated super app interface, thus positively affecting both direct and indirect engagement. Hence, the conducive presence of physical, financial, technological, legal, social, and cultural landscapes in emerging markets will likely enhance direct and indirect user engagement with the super app if they adopt one. Thus, we propose:

P3. The design thinking AI-embedded super app showcases a higher level of (a) direct and (b) indirect customer engagement in emerging markets compared to developed markets.

Network co-dependence refers to the inter-dependence of users within a social system, where peer behaviors and collective expectations shape individuals' decisions. This dynamic plays a significant role in indirect engagement with super apps. Customers often rely on recommendations and usage patterns of their peers. As one user noted, "Everybody, including local vendors, is using the app. I have to install it to keep up in society" (M, 24 years). This behavior reflects community homophily, where shared identity and usage create self-reinforcing adoption cycles (McPherson et al. 2001; Zhang et al. 2017).

Generative AI embedded in super apps amplifies the effect of these social dynamics by delivering network-based recommendations, personalized offers, and referral incentives. As users adopt the platform, their engagement generates new data, which in turn improves the experiences of others within their network (Markus 1987; McPherson et al. 2001). Network co-dependence also triggers critical mass dynamics, where adoption increases rapidly once a tipping point is reached. This has been particularly visible in communication and social media services that form the basis for super app expansion (Markus 1987; Salehi et al. 2023). In such environments, firms benefit from stronger indirect customer engagement, driven by trust, familiarity, and perceived social obligation. Therefore, we propose:

P4. The design-thinking AI-embedded super app showcases higher indirect customer engagement in network co-dependence than its effect on direct customer engagement.

The literature further highlights that value co-creation increases as engagement deepens (Kumar et al. 2019). These moderating variables, market context and network co-dependence, offer critical insight into how super apps must tailor their design and strategy for optimal adoption.

In the following section, we examine how engagement in super apps creates tangible and intangible value for multiple stakeholders.

5.5 | Value Creation for Stakeholders

Customer engagement is the battleground for firms' dominance to mobilize the business efficiently (Beckers et al. 2014). Through the lens of the engagement theory for value creation, in the context of the super app, engagement with the app may generate value for the following stakeholders: the customers, the super app firm, and the participating service providers.

The derived value for customers can be comprehended as lifestyle enhancement, customer empowerment, and social inclusion. Using a super app ensures customers' *lifestyle enhancement*, providing easy and convenient solutions with AI-infused personalization for their everyday needs. One of the customers stated, "The diverse services available on this app really uplift my way of living. It's like having a digital assistant that helps me manage everything efficiently." (F, 42 years). This reflects lifestyle enhancement, whereby the app becomes embedded in routine behavior, offering time-saving and intuitive experiences. *Customer empowerment* showcases increased customer interaction with the firm and access to information for better decision-making for future buying, thus enhancing the experience and self-efficacy (Pansari and Kumar 2017). The super app has the potential to bring inclusivity by allowing customers of every segment to interact and transact with mainstream businesses. Hence, *social inclusion* depicts the relational value of society and indulgence in similar habits of dominant societal trends (Markus 1987). As one interviewee confirmed, "This one super app helps me fit in socially because everyone else is using it too. I feel like I am part of a larger community." (M, 24 years). This relational value, the sense of being connected to societal norms and shared habits, contributes to both functional and emotional loyalty.

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The *derived value of the super app firm* can be delineated in terms of increased profitability, app sustainability, and customer opt-in. Like any business, the super app firm would intend to achieve a market share, leading to *increased profitability*. Accordingly, the platform will plan and strategize for marketing activities to leverage the engagement further. Furthermore, access to products/services/verticals under one umbrella app will likely ensure *app sustainability* with enhanced usage. One of the interviews underscored, "Having business through apps is essential for businesses. It is hard for them to sustain one app as it requires a workforce. Super apps have cashed this problem of small businesses to achieve long-term engagement." (M, 46 years). Super app firms continuously engage with customers through technologies like generative AI and ML, providing experiential usage to achieve a sustainable business model (Lee 2018). It is expected that high usage, that is, direct engagement with the app, will lead to *customers opting in and being* willing to sign up to receive the firm's marketing material (Pansari and Kumar 2017).

The *derived value for participating firms* can be defined in terms of market visibility, increased market share, and better return on investment. Collaborating with the super app provides participating firms access to a broader customer base against standalone mobile apps (Flora et al. 2014). Super apps use centralized promotions and customer-serving experience *to* improve *market visibility* for participating firms. As a participating firm manager acknowledged, "Customers download our mini app version through the super app, and we can widen the market base." (F, 39 years). This extended reach and exposure lead to positive word-of-mouth (WOM), referrals, and more robust customer feedback loops, ultimately driving higher revenue and *increased market share*. Importantly, indirect customer engagement, such as peer recommendations and content sharing, plays a larger role in delivering value to participating firms than direct engagement, given their dependence on network externalities and shared usage behavior (Skrt and Antoncic 2004). Therefore, the participating firms will likely have *better returns on investment* by receiving economies of scale, leading to a symbiotic relationship between participating firms and the super app. Hence, we propose that.

P5. The (a) customer and (b) super app firms derive high value when the customer is, both directly and indirectly, engaged with the app; however, for (c) participating firms, higher value is derived when customers are more indirectly engaged.

6 | Bringing a Super App to the Marketplace Success

6.1 | Managerial Implications

The proposed process framework has widespread implications for managers of super app firms, participating firms, and government policymakers, providing them with a blueprint for designing a human-centric digital product to achieve value-driven customer engagement.

At its core, the framework guides balancing design thinking elements: desirability, feasibility, and viability, to support adoption and usage. Successful super apps will require iterative refinement of features using real-time user data to deliver both functional value and emotional resonance. Managers in emerging markets can especially benefit by leveraging mobile leapfrogging and high digital engagement to scale adoption quickly. In networked societies with strong peer-to-peer influence, indirect engagement mechanisms, such as referrals and community sharing, should be strategically prioritized.

To support implementation, we introduce a 2 × 3 strategic matrix to help super app managers plan their marketing activities to generate more engagement value, as shown in Table 6. The level of market penetration in the population—high, moderate, and low—is represented on the *X*-axis. The stakeholder categories to be engaged with—customers and participating firms—are represented on the *Y*-axis. This strategic matrix provides guidance on specific implementable strategic activities to further engage with its stakeholders, categorized as *sustenance, growth, and acquisition*. The suggestive strategy depends on to what extent the firm has implemented the customer engagement process.

TABLE 6 | Strategy matrix for stakeholders of super apps.

Strategy for super apps to engage with the stakeholders[a]	Penetration percentage of mobile application/super apps →		
	Sustenance	Growth	Acquisition
	High (100%–75% population)	Moderate (75%–40% population)	Low (less than 40% population)
Customers	Run loyalty programs for existing customers; leverage AI to enhance the user interface and experience by continually evaluating existing customer journeys.	Provide more cashback offers to acquire more customers. Focus on the financial features of the application, such as credit systems/loan schemes.	Increased awareness by extending offers. Adding more AI-embedded features to the application with minimal cost.
Participating firms	Extend flexible revenue sharing model, incentives to launch innovative products/services and pull customer traffic.	Mergers and acquisitions with the SMEs to bring immediate traffic for a particular geography and product category.	Developing an efficient payment/wallet system with AI-supported security encryption. Generating awareness for the platform.

[a]The strategy of super apps depends on the extent to which the customer engagement process is implemented and achieved.

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TABLE 7 | Research agenda.

S. No.	Basis	Research question	Rationale for further exploration
RQ1	Super app firm related	How can we empirically validate, measure, and quantify the weightage of factors in the super app framework?	It is vital to examine the magnitude of the impact of each factor on the design and diffusion of the super app. Future studies can extend the present framework by conducting empirical studies examining each construct of the design thinking approach.
RQ2		How can a super app have a unique value proposition with generative AI to gain a competitive edge over other super apps?	Firms can create a unique value proposition through generative AI in the business model. Researchers can further explore potential avenues for developing unique features and services to differentiate and gain a competitive edge for their super app.
RQ3		What forces affect a super app's acquisition and market share?	It is crucial to examine whether any forces and factors impact how super app acquires participating firms' services. Researchers must examine any distinct variability in the market share due to the acquisition of participating firms.
RQ4		How can the use of AI technology in super app firm benefit from the customer data and information to its best advantage?	The data collated by the super apps are shared with the participating firms to curate personalized content using AI. Balance is needed to maintain a win-win situation for the super app and stakeholders leveraging data, that is, participating firms and customers.
RQ5	Participating firm related	How does AI impact the user engagement of mini apps within super apps impact compared to standalone apps?	Participating firms might have distinctive AI features and services as a mini app under super app compared to their independent app providing similar services. Establishing the essence of having the same app in two identical or different formats is essential.
RQ6		What factors are the participating firms considering while deciding to become part of the super app?	Several legal, financial, and operational consequences exist for the participating firms acquired under the super app. Further studies need to focus on the considerations of the participating firms to maintain their visibility while becoming a part of the super app.
RQ7		How can the native participating apps influence the diffusion of a super app in other markets?	The study highlights the interdependent relationship between super app firm and participating firms. Researchers must explore how super app adoption can be further extended to other markets through the international expansion of native mini apps.
RQ8	Customer related	What role will smart devices and new-age technologies play along with generative AI in enhancing super app engagement?	Super apps pair up and experiment with evolving new-age technologies such as Generative AI, AR, VR, ML, and futuristic accessories like smart wearable devices to multiply data and information.
RQ9		Does indirect customer engagement with the super app bring more value to customers than direct engagement? Are there any cons of using super app, and how does it impact value creation?	Indirect customer engagement enhances the customer's status and connection with the firm. Customers receive value for their feedback, referrals, and WOM. It is interesting to explore the differential impact of indirect customer engagement relative to the satisfaction from direct engagement. There might be a negative impact of using super apps for the customers that influences the adoption and value creation.

For example, in markets where the super app already achieves high penetration (75%–100%), firms should adopt a *sustenance*. For customers, this may include AI-powered loyalty programs, gamification, or personalized rewards to increase indirect engagement and brand affinity. For participating firms, direct engagement can be strengthened via flexible revenue-sharing models or bundled analytics support, while indirect engagement can be increased through AI-driven service innovation that encourages continued collaboration.

In addition, super apps' metadata collection capabilities enable marketers to make informed, real-time decisions across service design and customer experience. However, these capabilities also carry significant risk, as any breach of customer data

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privacy can quickly undermine user trust. Therefore, managers must implement robust data protection protocols, including encryption, consent-based usage, and constant system monitoring to mitigate threats of hacking or data misuse.

6.2 | Academic Implications

To the best of our knowledge, this is the first comprehensive study to explore the super app phenomenon within marketing literature using a human-centered design and engagement lens, offering a robust conceptual framework grounded in thematic analysis. By grounding our work in human-centered design thinking literature, we extend its elements—desirability, viability, and feasibility—to address the complexities of an AI-embedded, multi-stakeholder ecosystem of super apps (Meinel et al. 2020; Seidel and Fixson 2013). This positions design thinking not merely as a tool for customer-centric innovation but as a strategic framework for navigating the innovation and user-centricity in dynamic digital environments.

We propose a human-centered construct for super apps, characterized by six core AI-embedded attributes: multi-purpose single platform, mediation platform, integrated digital payments, inclusivity, individual app environments, and optimized functionality. These attributes address key gaps in traditional apps while contributing to mobile app and innovation literature.

Our study advances debates on customer engagement and value co-creation by highlighting how super apps leverage generative AI to enhance both direct experiences (e.g., personalization, seamless navigation) and indirect mechanisms (e.g., referrals, social influence) (Glikson and Woolley 2020). This broadens the scope of customer engagement literature, framing super apps as platforms for multi-stakeholder value creation (Kumar and Pansari 2016).

Furthermore, the framework introduces two critical moderators for super app adoption, market context (emerging vs. developed countries) and network co-dependence, that shape super app adoption and its impact on customer engagement. By doing so, it advances both the customer engagement and super app literature. In addition, the framework highlights the types of value generated through super app adoption for key stakeholders, including the super app firm, participating firms, and customers. These insights underscore the importance of our model in structuring and expanding the customer engagement process within the context of AI-enabled digital innovation (Dinner et al. 2015; Hehn and Uebernickel 2018).

7 | Future Research

This study has developed a process framework that explains the factors influencing super app development and the impact on related stakeholders. Furthermore, we present the following future research agenda in the super app domain, categorized into three broad areas (Table 7).

First, we present four research questions regarding the super app firm and innovation for an in-depth investigation of the process framework. These include the empirical validity of the suggested measures for the super app framework (RQ1), the unique value proposition with AI for the competitive edge of the super apps (RQ2), forces affecting the acquisition and market share of the participating firms (RQ3), and customer information leveraged to the advantage of the super app firm (RQ4). Second, we present three research questions for the participating firms under super apps. These include differentiated generative AI features of participating firms from their standalone app (RQ5), factors affecting business decisions for participating firms partnering with a super app (RQ6), and the adoption of super apps in developed markets with the expansion of native participating firms (RQ7). Third, we present two research questions emphasizing the customer to enhance the value derived from the super app. They include the role of smart wearable devices and new-age technologies like generative AI in enhancing super app features for better customer engagement (RQ8) and the impact on value creation with indirect customer engagement rather than direct customer engagement, as well as the cons of a super app (RQ9).

8 | Conclusion

The rise of super apps has undeniably ushered in a new era of convenience and interconnectedness, reshaping how users interact with mobile platforms. While the potential of these apps is immense, their continued success depends on addressing critical challenges related to privacy, data security, and ethical design. These issues must be mitigated to build widespread global trust and adoption. Amidst these constraints, generative AI has become a strategic imperative, offering the ability to personalize experiences, enhance security, and streamline engagement. When combined with a human-centered design thinking approach, super apps can be developed not only to meet user needs but to foster long-term trust, innovation, and sustainability (Hehn and Uebernickel 2018). The demand for super apps is particularly evident in the dynamic regions across Southeast Asian countries such as Japan, Indonesia, South Korea, Singapore, Dubai, and India, which already have a potential super app. In these markets, the super apps create meaningful synergies across stakeholders by empowering customers, expanding opportunities for participating firms, and delivering strategic value to platform providers. With continued innovation, the super app industry is well positioned to redefine digital engagement and shape the future of how we experience and interact with everyday services.

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Ethics Statement

The authors have read and agreed to the Committee on Publication Ethics (COPE) international standards for authors.

Conflicts of Interest

The authors declare no conflicts of interest.

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Endnotes

[1] We define the 'super app' as a mega aggregator that hosts a range of service providers (also referred as participating firms). Throughout the article, we use the terms '*super app platform*' and '*super app firm*' interchangeably, depending on context. Similarly, '*mini apps*' and *'service providers'* are used interchangeably to refer to participating firms.

[2] MSCI classifies markets and countries based on the economic development, index stability, and market accessibility criteria. Emerging market countries are those exhibiting low to moderate on these criteria while the developed market countries meet these requirements at a high level (https://www.msci.com/our-solutions/indexes/market-classification). Emerging markets include Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Korea, Kuwait, Malaysia, Mexico, Peru, Philippines, Poland, Qatar, Saudi Arabia, South Africa, Taiwan, Thailand, Turkey, United Arab Emirates. We have used 'emerging market' and 'emerging country' interchangeably.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Web appendix (WA). **WA-A1:** Comparison of super apps versus other traditional apps. **WA-A2:** Literature review on super app and proposed super app characteristics. **WA-A3:** Marketplace evidence for super apps and associated insights. **WA-A4:** Description of interviews participants. **WA-A5:** List of super apps with active users.

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Dynamic AI-embedded Super App: A Design-Based Process Innovation for Customer Engagement and Value Creation

Web Appendix (WA)

WA-A1: Comparison of super apps vs. other traditional apps

Types of Apps	Traditional Standalone App	Multi-purpose App Suite	Super App
Definition	A single mobile application with a single main function exclusively (Osio 2019)	A collection of apps with related functions and the ability to integrate and often owned by the same company	A single hybrid app with seamlessly integrated 'mini programs' within the app, facilitated by a payment system by the aggregate app company
Focus	Singularly specific and limited functionalities	A related and similar set of functions with a similar purpose	Diverse utility on-demand functions, specifically entertainment, digital payment, e-commerce, food delivery, ride booking
Input / Resources	- Technical support - Basic IT infrastructure - Can be specific to an intended operative system: iOS or Android	- Established customer user base - Brand expertise in a set of functionalities - Interoperability based on single login	- Technical expertise - Cross-platform compatibility - Advanced AI-tech infrastructure - New customer bases via partnerships - Financial strength & investment
User Interface	One app focusing mainly and exclusively on one service present in the main app system	A bundle of fragmented apps with similar use features collated in one app suite	Individual app environments of diversified mini-programs seamlessly integrated within the super app ecosystem of on-demand features
Value Created	- Customer retention - Specific service-oriented branding - Built-in features	- Customer trust in expertise - Expertise-oriented branding - Bundled features	- Customer engagement - Experience-oriented branding - Personalization with on-demand features
Evidence	Example: Twitter, Skype, Spotify, Candy Crush	Examples: Google Suite, Microsoft 365, Adobe Creative Suite	Example: WeChat, Alipay, Baidu, Meituan, Tmall

WA-A2: Literature Review on Super App and Proposed Super App Characteristics

Section 1: Literature Review Findings						Section 2: Super App Characteristics					
Author (s)	**Context of Study**	**Method**	**Market Context**	**Factors for Super app Development**	**Findings**	**Digital Payment Mechanism**	**Mediation Platform**	**Inclusivity**	**Multi-purpose Single Platform**	**Individual App Environment**	**Optimized Functionality**
Nayebi and Ruhe 2017	Optimized super app functions	Exploratory Study	Global	- Cohesiveness - Feature extraction and value - Effort Estimation - User Feedback	- An app is considered valuable when features are cohesive with co-occurrence of existing apps. - Cohesiveness ensures semantic connections across features are maintained and optimized.			✓	✓	✓	✓
Carvalho Ota et al. 2020	Super app and privacy	Exploratory Study	Developed Market (Europe)	- Mobile user interface - Privacy concerns - Scalability	- The super app avoids the need for reinstallation every time a member app adds a new feature.			✓			✓
Cheng et al. 2020	Implication of super app innovation on mobile platform ecosystem	Exploratory Study	Emerging Market (China)	- Customer Access - Distribution - Portability - Development costs -Hardware -software - Internet access	- The native apps do not meet the specific needs of users and can be supplemented by mini-programs. - Super app fills the gaps in traditional apps by reducing fragmentation and app development cost	✓	✓		✓	✓	✓
Fasnacht 2021	Super ap and banking	Exploratory Study	Emerging Market (China)	- Agility - Financial strength - Partnerships - Technology - Socio-cultural factors	- Super app's digital ecosystems prioritize customers and data while consolidating interconnected commodities and services. - Agile management approach is needed to achieve sustainable financial growth, that builds upon partnerships to connect technologically, socially, and culturally.	✓			✓	✓	✓

Huang and Miao 2020	Micro-behavioural process of super app	Qualitative Study	Emerging Market (China)	- Re-domestication of technology - User privacy - Macro factors - Cultural customization	- Customers suspend the use of the super app periodically with a lack of engagement - Super app experience gets affected by prevailing social norms and rituals. - The pattern of using and not using mobile apps changes in daily life, and this hidden habit may be a regular way people manage their social networking through phones.		✓	✓	✓	✓	✓
Roa et al. 2020	Impact of alternative data from super app	Empirical Study	Developed Market (Latin America)	- Customer demographics - Financial services - Type of data extracted	- Super-app diversity is attributable to its mini-program function - Super app can be quite helpful for financial institutions	✓	✓		✓	✓	✓
Steinberg 2020	Cultural Platform-ization of for super app	Descriptive Study	Emerging Market (East Asia)	- Platformization - Geo-cultural specificity - Revenue model - Mediation Platforms	- Super apps encourage new producers to reach new markets - The value proposition of a culturally customized super app is its mutual dialogue with customers.	✓	✓	✓	✓		
Zhu et al. 2023	Drivers of service extension success for a super app	Empirical Study	Developed Market (Japan)	- Trust - Habit - User attitude - Intention to use - Perceived fit	- Users who trust and frequently use the main app (LINE) are more likely to view its new service (LINE Pay) positively, especially if it is perceived to be a natural fit. When the core app and new service work well together, users develop a stronger positive attitude. - Privacy concerns make users less likely to have positive attitude for super apps				✓		✓
Present Study	Design thinking embedded super app customer engagement	Triangulation Method with Thematic Analysis	Emerging Markets	- Viability - Desirability - Feasibility - Moderating Variables -Value Creation for Stakeholders	- Conceptualizing super app and dimensions -Understanding the driving forces and factors for designing a super app, moderating factors influencing customer engagement, and deriving value for the stakeholders.	✓	✓	✓	✓	✓	✓

WA-A3: Marketplace Evidence for Super Apps and Associated Insights

Super App (Market Base)	Super App Positioning	Factors for sustained engagement	Evidence
KakaoTalk by Kakao (South Korea)	Optimized app features	- Technology & infrastructure - Strong user base - Contextual Knowledge	KakaoTalk, Korea-based super app, has integrated AI to launch features such as summarizing unread messages and changing the tone of messages in chat rooms based on a lightweight language model (Byung-yeul 2024).
PayTM (India)	Secure digital payment mechanism	- Technology & infrastructure - Privacy & security - Skilled expertise	PayTM, India-based financial super app, introduced Paytm Intelligence (Pi), leveraging AI and ML to flag possible risks and frauds. It connect data signals from across the customer journey for digital safety and security (PayTM 2022).
WeChat by Tencent (China, Europe, US)	Serve all diverse services of daily life providing inclusivity and customised services	- Established customer base - Local market context - Evolving consumption pattern - Partnerships & mergers	WeChat, China-based super app, uses generative AI to improve video games content and for social media ad recommendations. Tencent built research centres for practical AI applications for business in content, online gaming, social, and cloud services (Marr 2018, Deng and Qu 2023).
GoJek (Indonesia, Vietnam)	Leverage technology and partnership for one-stop solution	- Customer traction - Partnership & mergers - Enhance market reach	Gojek, an Indonesia-based super app, has partnered with TenMax.io to provide AI techniques for one-stop advertising solution to plan, measure, and optimise digital marketing efforts across social media and content sites (Chevi 2021).
Alibaba by Taobao & Tmall (China)	Combine features with mediation services	- Technology & infrastructure - Skilled expertise - Gain competitive edge	Alibaba, China-based super app, provides ML and cloud-based solution for developers with a rich ecosystem, including components for app infrastructure and business features, containers for running mini apps, and services for DevOps and operations (Hopkin 2023)
M-Pesa (Kenya)	Virtual marketplace for various products and services	- Subscriber base - Emerging market ecosystem - Technology -Trust and access	Safaricom, a Kenya-based company, launched its M-Pesa super app, driven by mobile network operators that leverage local infrastructure and data of the smartphone-enabled customers. It improvises the app with customer feedback (Gilbert 2021)

Super App (Market Base)	Super App Positioning	Factors for sustained engagement	Evidence
Careem (Dubai)	Multi-purpose single platform	- Personalization - Localized experience - Privacy and security - AI skilled expertise - Partnerships & Collaborations	Careem, Dubai-based ride hailing super app, is using AI to improve customer and driver security through targeted facial-recognition checks of drivers and latest estimated times of arrival to customers (ETAs) based on geographic and cultural factors. Careem educates engineers and business counterparts on AI and collaborates with academic institutions for AI innovation (McKinsey 2022).
LINE (Japan, Taiwan, US)	Comprehensive services driven by communication	- Local market knowledge - Lack of hardware availability - Trust with data security	LINE, messenger app to super app platform, added cultural and economic activities (DMFA 2020). Technically incompatible with android phones and restrict data access for social commerce in the US (Gundlapalli 2021)
Tata Super App (India)	Platform for connecting and rewarding customers experiences	- Financially strong firm - Strong partnerships - Stickiness via loyalty rewards - Community network effects	Tata Super app, India-based upcoming super app, introduced Neu coins for integrated payment system, i.e., a currency that allows transactions. Backed by industry giants, Tata, super app is has tied up with global conglomerate brands. (Balachandran 2022
Grab (Malaysia, Indonesia, Vietnam)	Digital empowerment by enabling hyperlocal consumer	- Customer consumption - Deep market knowledge - Infrastructural leverage - Strong brand reputation	Grab, focuses on hyperlocal user retention and market share in densely urbanized regions with transportation challenges. It scaled regionally, serving highly used services like ride-hailing, food delivery, and e-payments (Marketing week 2020)
Meu Carrefour by Carrefour (Brazil)	Building an inclusive platform for diverse services	- Personalized services - Customer engagement - AI technology	Meu Carrefour, Carrefour Brazil's super app, introduced a loyalty program called "My Rewards" which offered AI-based virtual coins for purchases, convertible into discount coupons and personalized recommendations (Mancuso et al. 2023).
Avo by Nedbank (Africa)	Digital payment mechanism with diverse services	- Customer base - Security and privacy - Evolving customer needs	Avo, an AI-embedded super app by Nedbank, offers a comprehensive platform that tailors services to customers' lifestyles. AI ensures bank-grade security and facilitates secure payments for enhanced user experience (Nedbank 2023, TechCentral 2020).

WA-A4: Description of Interviews Participants

Countries	Asia	North America	Europe and Middle east
Super app firms and Participating firms			
Interviewees role (s)	Executive (3), Project Manager (2), Product Design Manager (2), UI Architecture (1)	Executive (2), Project Manager (1), Strategy Manager (1), Director (1)	Executive (2), Vice President (1), Product manager (1), Marketing Manager (1)
Total Interviews	17	5	7
Data collection Period	Dec 2019-Feb-2022	Mar 2020-April 2022	May 2020-Jun 2022
Customers			
Age group (s)	Age 18-30 (4), Age 30-45 (3), Age 45 and above (2)	Age 18-30 (3), Age 30-45 (3), Age 45 and above (2)	Age 18-30 (2), Age 30-45 (1), Age 45 and above (1)
Total Interviews	12	15	14
Data collection Period	Jul 2020-Sep-2022	Oct 2020-Feb 2022	Mar 2021-Apr 2022

WA-A5: List of Super Apps with Active User

Country	Primary Industry	Super App Name Existing/Potential	Monthly Active Users[1][2][3][4][5][6][7] across countries	Monthly Active users in the Country of Origin[8][9][10]	Internet Users Population Absolute (Mobile Internet Percentage)[11][12]	Percentage population using the application
China	Telecommunication	WeChat (Tencent)	1.24 billion+	1.05 billion	1.05 billion (99%)	85%
China	Online Food Delivery	Meituan-Dianping	339 million+	339 million	1.05 billion (99%)	32%
China	E-Commerce	Alipay	652 million+	652 million	1.05 billion (99%)	62%
India	Mobile Payment	Paytm	300 million+	300 million	692 million (82%)	43%
Indonesia	Courier Delivery and - hailing services	Go-Jek	38 million+	29 million	213 million (94%)	14%
Singapore	Transport Services	Grab	32 million+	4.8 million	5.4 million (99%)	89%
Vietnam	Telecommunication	ZaloApp	60 million+	60 million	77.9 million (95%)	77%
South Korea	Telecommunication	Kakao	53 million+	47 million	50.6 million (97%)	93%
Colombia	Delivery Service	Rappi	10 million+	9 million	39.8 million (72%)	23%
Japan	Telecommunication	Line	178 million+	92 million	102.5 million (97%)	90%
Russia	Technology	Yandex Go	88 million+	62 million	127.6 million (86%)	49%

[1] https://www.businessofapps.com/data/app-data/
[2] https://paytm.com/document/ir/annual-reports/annual-report-2021-22.pdf
[3] https://expandedramblings.com/index.php/go-jek-statistics-and-facts/
[4] https://restofworld.org/stat-of-the-day/vietnam-everything-app/
[5] https://www.statista.com/statistics/746249/south-korea-kakaotalk-monthly-active-users/
[6] https://www.businessofapps.com/data/line-statistics/
[7] https://www.statista.com/topics/10280/yandex/
[8] https://www.statista.com/statistics/1032630/china-leading-apps-by-monthly-active-users/
[9] https://expandedramblings.com/index.php/grab-facts-statistics/
[10] https://www.rademergent.com/p/rappi-latams-super-app
[11] https://www.statista.com/statistics/262966/number-of-internet-users-in-selected-countries/
[12] https://www.statista.com/topics/779/mobile-internet/